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System Requirement Specification

SpaceB

Redesign Charging Unit Battery-Pack SP (NIMH)

PIDEU - Model

EUMBB00001M-C

Version: 2.0

Status: Release

Author: Denise Cappel

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Signatures

All signing persons affirm that the requirements specified in this document describe the system to be developed to its full extend. At the time of signature, no requirements other than those specified in this document are known.

No requirements other than those specified in this document apply for the system to be developed.

Additional requirements or changes to the requirements specified in this document require written form.

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Change History

Version	Date	Prepared by	Changes *- related Jama baseline -*
0.1	2021-11-04	D. Cappel	First export of System Requirement Specification *SysReqSpec_0_1*
0.2	2021-12-07	D. Cappel	Partial integration of customer's review comments *SysReqSpec_0_2*
1.0	2022-01-24	D. Cappel	Unchanged, version number was adapted to customer version counting *No new baseline*
1.1	2022-02-03	D. Cappel	Initial Version, Update of several threshold values *SysReqSpec_1_1*
1.2	2022-02-04	D. Cappel	Update of threshold values, signed version *SysReqSpec_1_2*
2.0	2022-05-11	D. Cappel	Update of threshold values, release - signed version *SysReqSpec_2_0*

Remark
A detailed change history can be generated from baseline comparison on demand.

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1 [BB000-SET-9] System Requirements

Upstream Customer IDs:

1.1 Introduction

1.1.1 Purpose

This document describes the fundamental system requirements of product development. The purpose of this document is to provide a system overview, specify the interfaces and list all functional and non-functional requirements that are essential to the success of the development of this product.

1.1.2 Requirement Wording Convention

The key words “shall”, “shall not”, “should”, “should not”, “may” and “may not” inside this specification are to be interpreted as follows:

Shall: This word means that the definition is an absolute requirement and therefore mandatory.

Shall not: This means that the definition is an absolute prohibition.

Should: This word means that the definition is a recommendation, which shall be followed, unless good reasons are stated for not doing so.

Should not: This means that the definition is a dissuasion, which shall be followed, unless good reasons are stated for not doing so.

May (not): This means that the definition is a permission, which can be followed, but does not have to, meaning it’s optional.

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1.1.3 System Overview

The system to be developed is a charging unit (CU) for a nickel metal hydride (NiMH) battery pack.
The battery cell-stack is not part of the system (CU).

The battery pack will be used as a mains-independent power supply in different infusion pumps.
The battery pack can additionally be charged in a charging tray.

The CU monitors

- the temperature of the battery cells;
- the voltage of the battery cell-stack;
- the discharging current (+Vout) and the charging current.

The CU monitors and controls the charging process of the battery cells.

The CU provides a power interface (+Vout, +Vin, GND) and a communication interface (TxD, RxD).
Power has to be supplied via pin +Vin for charging.
For discharging, power from the battery pack is provided via pin +Vout.
The communication interface allows to read out different status-bits, errors and an estimated state of charge.

The following simplified block diagram gives an overview of the physical architecture and the boundary of the system (CU).

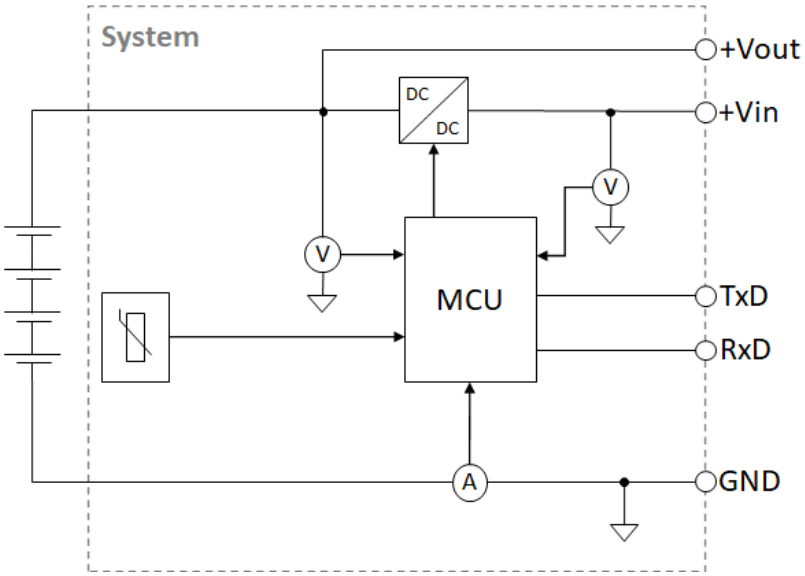


Figure: Simplified Block Diagram of the battery pack (cell stack + CU)

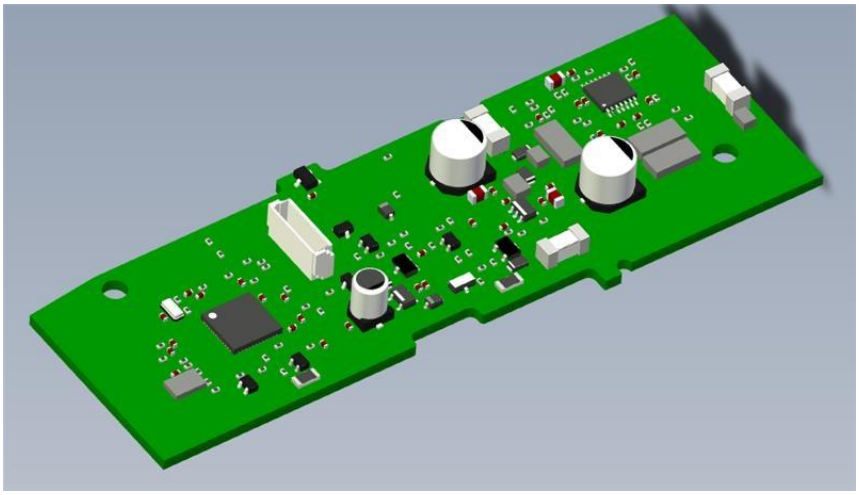


Figure: Example for the physical CU

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1.1.4 References

This list of references includes different documents and/or publications referenced inside this specification.
The referenced documents and/or publications in this list do not incorporate additional requirements unless explicitly stated inside this specification and may solely serve an informational purpose.

[CRS] B. Braun Melsungen AG, Hostpital Care Division, CoE Automated Infusion Systems, *M001_000771, Component Requirement Specification (CRS), Redesign Charging Unit Battery-Pack SP (NIMH),* Version 3.0, 2022-04-13
(M001_000771_Space Component Requirement Specification NiMH Battery - Charging Unit V3.0.pdf)

[DIA] B. Braun Melsungen AG, Division Hostpital Care, *M001_000776, Documentation Matrix, Redesign SW-Charging Unit of Battery-Pack SP (NiMH),* Version 1.0, 2021-02-05
(DE-2021-0015230-Entwicklungsvertrag Panasonic - Annex Technical Document....pdf)

[BSP] Panasonic Industrial Devices Europe GmbH, *BB000-SDS-01, Bootloader Specification (SpaceB BMS),* Version 1.0, 2022-04-19
(BB000-SDS-01-Bootloader_Specification.pdf)

1.1.5 Acronyms and Abbreviations

CU	Charging unit
EE	Estimation error
FCC	Full charge capacity
NiMH	Nickel metal hydride
NTC	Negative temperature coefficient thermistor
PCB	Printed circuit board
RC	Remaining capacity
RR	Remaining runtime
RxD	Receive data
TxD	Transmit data

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1.1.6 Project Contacts

1.1.6.1 Panasonic Contact Information

The contractor, hereafter termed *supplier*, is by name:

Panasonic Industrial Devices Europe GmbH
Zeppelinstraße 19
21337 Lüneburg
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Role	Name	Mail
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Product Manager (Marketing)	Neumann, Philipp	philipp.neumann@eu.panasonic.com
Software Project Manager (SPM)	Winkelmann, Christian	christian.winkelmann@eu.panasonic.com
Hardware Project Manager (HPM)	Pervaiz, Usman	usman.pervaiz@eu.panasonic.com
Requirements Engineer (ReqENG)	Cappel, Denise	denise.cappel@eu.panasonic.com

1.1.6.2 Customer Contact Information

The ordering party, hereafter termed *customer*, is by name:

B. Braun Avitum AG
Schwarzenberger Weg 73-79
34212 Melsungen
(Germany)

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1.2 (External) Interfaces

1.2.1 Electrical Connectors

1.2.1.1 External Connector

1.2.1.1.1 [BB000-SysReq-10] External Connector

The CU shall provide an electronic interface with the following pin assignment.

Pin	Name	Description
1	+Vin	Power supply input voltage
2	RxD	Receive data
3	GND	Ground
4	TxD	Transmit data
5	+Vout	CU output voltage

The electronic interface shall be implemented by empty pads on the PCB as specified in Mechanical Dimensions

Remarks:
The following figure is given as a visualization of the pin assignment, when the CU is assembled in the battery-pack housing by the customer.



Figure: External connector of housing provided by the customer (schematic view from outside), [CRS, ID 9]

Upstream Customer IDs:
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1.2.1.1.2 [BB000-SysReq-37] Communication Interface

The CU shall provide a communication interface through the pins TxD and RxD.

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The communication shall be of type master-slave and realized by a UART, for operation inside the pump. The CU shall drive an LED through pin TxD, for operation in the external charging tray, to indicate different charging conditions.

Remarks:
For interface/UART configuration refer to Communication Interface Configuration and Physical Communication Interface
For communication protocol refer to Communication Protocol
For LED indication refer to External LED Control

Upstream Customer IDs:
46

1.2.1.2 Battery Connector

1.2.1.2.1 [BB000-SysReq-11] Battery Connector

The CU shall provide a connector for the battery cell stack.
The connector shall be implemented by empty pads on the PCB as specified in Mechanical Dimensions and Pad Size of Connectors

Remarks:
The battery cell stack itself is beyond the system boundary of the CU.
The battery cell stack will consist of four NiMH battery cells in series of type Panasonic HHR-210AH (Cylindrical design (diameter 17 mm, height 50 mm)).

Upstream Customer IDs:
10

1.2.2 Other Interfaces

1.2.2.1 [BB000-SysReq-12] Hall Effect Sensor

The CU shall incorporate a Hall-effect sensor.
The sensor shall be of the following type.

Type
Infineon TLE4913

Remarks:
The sensor detects the correct fixing of the battery cover (magnet) in the pump.

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The output signal state of the sensor shall be provided over the communication interface, see Magnetic Field Sensor State
For position of the sensor refer to Mechanical Dimensions.

Upstream Customer IDs:
11

1.3 Functional Requirements

1.3.1 System States

1.3.1.1 [BB000-SysReq-85] Active Mode

The CU shall be in active mode, if it is not in standby or in shutdown.

Within active mode, the CU shall distinguish between charging mode and discharging mode.

The CU shall be in charging mode, if all of the following conditions are valid.

Condition	Comment/Reference
Power supply detected as connected	Power Supply Detection
No unallowed discharging	Unallowed Discharging
Set-to-discharge command inactive	0x64: Set to Discharge Mode

The CU shall be in discharging mode, if one of the following conditions is valid.

Condition	Comment/Reference
Power supply detected as disconnected	Power Supply Detection
Unallowed discharging detected	Unallowed Discharging
Set-to-discharge command active	0x64: Set to Discharge Mode

In charging mode, the CU shall charge the battery in accordance with the requirements of this specification.
In discharging mode, the CU shall not charge the battery.

Remarks:
In charging mode, there is not neccessarily a charging current flowing.

Upstream Customer IDs:

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1.3.1.2 [BB000-SysReq-58] Long-Term Storage

The CU shall detect a long-term storage, if all of the following conditions are valid.

Condition	Reference / Comment
Power supply disconnected	Power Supply Detection
No communication	Communication Detection
Measured discharge current < 30 mA	-

Subsequently, the CU shall evaluate the measured battery voltage and go to standby or shut down, as specified below.

Condition	Comment	Reference
Measured battery voltage > 4.4 V	Go to standby	Standby
Measured battery voltage <= 4.4 V	Shut down the electronic	Shutdown

Remarks:
Measured current refers to the current measured by the MCU corrected by the average self consumption of the electronic.

Upstream Customer IDs:
26

1.3.1.3 [BB000-SysReq-63] Standby

The CU shall reduce its energy consumption in standby, where the power supply is given by the battery only.
The CU shall alternate sleep and wake phases to achieve the self-consumption limits given in Self-Consumption Limits.

Furthermore, the CU shall detect the following conditions during the wake phase.
If one of the conditions is valid, the CU shall leave standby and enter active mode again.

Condition	Reference / Comment
Power supply connected	Power Supply Detection
Communication reconnected	Communication Detection
Measured discharge current >= 30 mA	Only during wake phase

Remarks:
For conditions for shutdown or standby, refer to Long-Term Storage
For decrementing of the capacity counter in standby, refer to Capacity Counter
Measured current refers to the current measured by the MCU corrected by the average self consumption of the electronic.

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Upstream Customer IDs:
27

1.3.1.4 [BB000-SysReq-32] Shutdown

The CU shall provide the ability to disconnect itself from the battery voltage (shutdown), in order to minimize the self-consumption by the electronics.
Before a shutdown, the value of the last FCC shall be saved.
The capacity counter shall be set to zero and saved.
The EE (estimation error - described in Full Charge Capacity Estimation) shall be set to zero and saved.

If a supply voltage gets connected to the input, the CU shall start operating again.

Remarks:
Self consumption of the electronic in shut shutdown is specified in Self-Consumption Limits
Criteria for a shutdown are specified in Long-Term Storage.

Upstream Customer IDs:
26
16

1.3.1.5 [BB000-SysReq-90] System State After Production

The CU shall be in active mode after production.

Remark:
This implies, that the CU will go into standby soon after production, due to missing input voltage and communication signals.
Without connected input-voltage supply, the CU will eventually shut down, as specified in Long-Term Storage.

Upstream Customer IDs:

1.3.2 Battery Monitoring

1.3.2.1 General

1.3.2.1.1 [BB000-SysReq-13] Battery Deep Discharged

The CU shall detect and communicate (in reply to 0x73: Read Status) deep-discharged cells.

Condition	Bit in 0x73: Read Status	Comment
Measured battery voltage <= 3.9 V	d = 1	Deep discharge
Else	d = 0	No deep discharge

Upstream Customer IDs:
12

1.3.2.1.2 [BB000-SysReq-23] Battery Temperature Monitoring

The CU shall detect and communicate the following condition (in reply to 0x73: Read Status).

Condition	Bit in 0x73: Read Status	Comment
Measured temperature < 0 °C or > 55 °C	h = 1	Temperature not in valid region
Else	h = 0	Temperature in valid region

Remarks:
For selection of the charge mode depending on the measured temperature refer to Temperature Conditions

Upstream Customer IDs:
12

1.3.2.1.3 [BB000-SysReq-47] Discharge Over-Current

The CU shall detect and communicate (in reply to 0x73: Read Status) the following condition.

Condition	Bit in 0x73: Read Status	Comment
Discharge current > 2.55 A for > 20 sec	i = 1	Discharge over-current
Discharge current < 2.5 A	i = 0	No overcurrent, bit reset

The CU shall, when detecting a discharge over-current, set the capacity counter and the remaining runtime to zero.

Remarks:
Typical discharge currents by the pump are approx. 0.1 ... 2.5 A.
The pump may regularly discharge with currents of 3.5 A, but not longer than 15 sec at a time.
The CU does not need to measuere the exact value of discharge currents > 2.55 A, it only needs to detect, if a discharge current is > 2.55 for more than 15 sec.
A margin and hysteresis is included.
The old system included an additional margin of 5 sec, which is included in the specified value from above.

Upstream Customer IDs: 23

1.3.2.1.4 [BB000-SysReq-65] Battery Voltage Minimum

The CU shall detect and communicate (in reply to 0x73: Read Status), the following condition.

Condition	Bit in 0x73: Read Status	Comment
Measured battery voltage <= 4.0 V	m = 1	Battery volt. at minimum / fully discharged
Measured battery voltage > 4.0 V	m = 0	Battery voltage above minimum

Remarks:
This voltage threshold is regularly reached in Battery Maintenance.
It could also be reached, if the battery is further discharged despite a remaining runtime estimation of zero, and can be refered to in service or tests.

Upstream Customer IDs: 41

1.3.2.2 Battery Function Test

1.3.2.2.1 [BB000-SysReq-49] Battery Function Test - Charging

The CU shall detect and communicate (in reply to 0x73: Read Status) the following conditions, when charging is intended (charging converter and controller on (*)).

Condition	Bit in 0x73: Read Status	Comment
Measured battery voltage <= 2.5 V	r = 1	Battery under limit
Else	r = 0	Battery ok

Condition	Bit in 0x73: Read Status	Comment
Measured battery voltage >= 6.57 V	v = 1	Battery over limit (defect or open wire)
Else	v = 0	Battery ok

If the CU detects one of the conditions above, the capacity counter (and thus remaining capacity and remaining runtime) shall be set to zero.
Charging shall be interrupted for 20 sec.
Then, the CU shall restart the charging process (retry).

Remarks:
(*) This also refers to the on-phase in trickle or warm-up charging.

Upstream Customer IDs:
33
34
12

1.3.2.2.2 [BB000-SysReq-50] Battery Function Test - Trickle Charge

The CU shall detect and communicate the following conditions to the pump in reply to 0x61: Get Accutest Result.

Condition	Bit in 0x61: Get Accutest Result	Comment
Measured output current > 80 mA	i = 0	Current ok
Else	i = 1	Current to small

Condition	Bit in 0x61: Get Accutest Result	Comment
Measured battery voltage > 4.6 V	u = 0	Voltage ok
Else	u = 1	Voltage to small

Condition	Bit in 0x61: Get Accutest Result	Comment
Current ok and voltage ok	p = 1	Test passed
Else	p = 0	Test failed

Condition	Bit in 0x61: Get Accutest Result	Comment
Always	e = 1	Battery test is enabled

If the test failed (p == 0), the CU shall additionally set the remaining runtime and the capacity counter to zero.

After timeout of the discharge mode (0x64: Set to Discharge Mode), the CU shall restart the charging process again.

Remarks:
This battery test is triggered by the pump every 60 min as follows.
The pump evaluates bit c of 0x73: Read Status.
If the CU is in trickle charge mode (bit c = 0), the pump starts the following test procedure.
The pump sends a request to discharge the battery, which allows discharging despite connected mains (can be retriggered, refer to 0x64: Set to Discharge Mode).
The pump then connects a current sink to the "+Uout" and "GND" pins (100 mA +-20 mA are expected).
After > 500 ms, the pump sends a request to get measurement results from the CU (refer to 0x61: Get Accutest Result).

Upstream Customer IDs:
33 35

1.3.2.3 Battery State Estimation

1.3.2.3.1 [BB000-SysReq-42] Capacity Counter

The CU shall accumulate the electrical charge flowing into / out of the battery.
The accumulated value is being referred to as the capacity counter.

The initial value of the capacity counter after production shall be set to the initial estimate of the FCC (Full Charge Capacity).

When an end of charge is detected, the capacity counter shall be set to the new (corrected) FCC (Full Charge Capacity Estimation).
During Trickle Charge Mode, the capacity counter shall not be increased or decreased.
In Standby the capacity counter can not be decremented during the sleep phase.
Therefore, it shall be decremented by a reference value representing the average self-discharging rate (updated at least every 4 min).

Remarks:
Accumulating the electrical charge shall be done by approximating the integral of the current flowing into / out of the battery over time (also known as Coulomb Counting).
The CU does not differentiate between standby inside or outside the pump, thus the reference value accounts for current consumption of the pump aswell.
The pump has a current consumption of < 100 µA.
For reset of the capacity counter also refer to Discharge Over-Current, Cut-Off Voltage

Upstream Customer IDs:
17 26

1.3.2.3.2 [BB000-SysReq-60] Cut-Off Voltage

The CU shall detect the following condition.

Condition	Comment
Measured battery voltage <= 4.2V	Cut-off voltage

At this cut-off voltage a normal discharge procedure should stop, in order to leave an energy margin until the battery is fully discharged.
Thus, when the cut-off voltage is reached during discharging, the CU shall set the capacity counter and the remaining runtime to zero.

Remarks:
When the cut-off voltage is reached during discharging, the RR estimate has been too high, or discharging has continued despite the RR being zero.
The pump may not have raised a pre-alarm or alarm, because RR was too high.
In this case, the pump will withdraw 100 mA for another 10 min to give an alarm to the user.
For normal alarm conditions refer to remarks of Remaining Runtime Estimation.
Information about the RC and the RR are communicated to the pump as specified in Capacity Counter and Remaining Runtime Estimation.

Upstream Customer IDs:
39

1.3.2.3.3 [BB000-SysReq-68] Remaining Capacity Estimation

The CU shall estimate and communicate (in reply to 0x63: Read Capacity) the remaining capacity (RC) of the battery.

The RC estimate shall equal the capacity counter multiplied with the following capacity availability factor.

For different temperatures, the factor shall be be determined by inter- or extrapolation.

Measured Battery Temperature	Capacity Availability Factor
-10 °C	85 %
25 °C	100 %
60 °C	90 %

Additionally, the RC estimate shall be limited to
- 0 <= RC <= FCC - EE.

Remarks:
The actual available battery capacity is dependent on the battery temperature.
Also refer to Capacity Counter and Full Charge Capacity Estimation

Upstream Customer IDs:
13

1.3.2.3.4 [BB000-SysReq-44] Remaining Runtime Estimation

The CU shall estimate and communicate (in reply to 0x72: Read Remaining Time) the remaining runtime (RR) of the battery.

The estimate shall be based on the RC as follows.

- RR = filter(RC / max(measured discharge current, 100 mA))

The filter shall be an IIR1 filter with a time constant of tau = 25 sec and a sampling interval of 1 sec.

Remarks:
To avoid huge estimates, at least 100 mA are considered in the calculation of the RR.
The pump triggers an alarm indication to the user, based on the RR, RC and FCC.
When the remaining runtime (estimated by the pump) reaches 30 min, the pump gives a "pre alarm".
The pump gives an "end-alarm" for 3 min, 30 min after starting the "pre-alarm".
After starting the pre-alarm, the pump does not stop the alarm until the battery gets charged again, even if the remaining runtime provided by the CU would increase.
For reset of the capacity counter also refer to Discharge Over-Current, Cut-Off Voltage

Upstream Customer IDs:
28
43
17
29
30
31

1.3.2.3.5 [BB000-SysReq-67] Full Charge Capacity

The CU shall estimate the Full Charge Capacity (FCC) of the battery.
The initial estimate of the FCC after production shall be set to the following value.

Parameter	Value	Unit
Initial estimate of the FCC	1800	mAh

The estimate of the FCC may change during operation of the CU, but shall be limited to the following nominal value.

Parameter	Max.	Unit
Nominal FCC	1900	mAh

Remarks:
For FCC estimation refer to Full Charge Capacity Estimation.

Upstream Customer IDs:
36
37

1.3.2.3.6 [BB000-SysReq-70] Full Charge Capacity Estimation

The CU shall update the estimate of full charge capacity (FCC), when an end of charge has been detected, as follows.

Effective

- FCC = last_FCC - EE,

where last_FCC ist the FCC that has been determined at the last end-of-charge detection.

EE is the estimation error and determined as follows.

- The initial value of the EE after production shall be zero.
- When the CU detects the cut-off voltage, EE shall be set to the value of the capacity counter(*).
- When the CU detects an end of charge, EE shall be set to zero(**).

Remarks:
An estimation error can occur due to e.g. changes in cell chemistry or temperature.
A negative estimation error can only be corrected during battery maintenance, where the battery voltage always reaches the cut-off voltage.
Also refer to Battery Maintenance, Cut-Off Voltage.
Since the capacity counter is also being corrected to the new FCC at an end-of-charge detection (refer to Capacity Counter), the RC automatically gets corrected, too.
(*) Before setting the capacity counter to zero.
(**) After calculating FCC.

Upstream Customer IDs:

1.3.2.3.7 [BB000-SysReq-53] Day Counter

The CU shall count and communicate (in reply to 0x74: Read Day Counter) the days since the last discharge.
A day counter shall be incremented depending on the estimate of the remaining runtime and an 24 h timer.

Every 24 hours, the day counter shall be incremented by 1, if the estimated remaining runtime is > 30 min.
At a value of 255, the day counter shall stay at 255 instead of being further incremented.

If the estimate of the remaining runtime is <= 30 min, the cells are effectively discharged.
The CU shall reset the day counter and the 24 h timer to zero.

The initial day counter value after production shall be 0.

Upstream Customer IDs:
41
40

1.3.2.3.8 [BB000-SysReq-61] Battery Maintenance

The CU shall detect and communicate (in reply to 0x64: Set to Discharge Mode with byte 3 = 0xA5), the following conditions.

Effective

Condition	Bit in 0x64: Set to Discharge Mode	Comment
Always in reply to 0x64: Set to Discharge Mode	s = 0	Charging disabled via serial command

Condition	Bit in 0x64: Set to Discharge Mode	Comment
Measured battery voltage > 4.0 V	a = 1	Maintenance running
Measured battery voltage <= 4.0 V	a = 0	Maintenance finished

Furthermore, the CU shall set the day-counter threshold (Day Counter) at the cut-off voltage during maintenance (Cut-Off Voltage).

Condition	Consequence
Measured battery voltage <= cut-off voltage	Day counter = 0

Remarks:
The purpose of battery maintenance is to prolong the life of the cells and to determine the available full charge capacity.
If the day counter is > 0 and battery full (>95%, estimated by the pump), the pump may indicate a required maintenance, which then needs to be triggered by the user.
The pump then requests discharging for maintenance (0x64: Set to Discharge Mode with byte 3 = 0xA5).
If the battery is fully charged (bit c = 0 in reply of CU), the pump discharges the battery.
The pump resends the discharge request (with byte 3 = 0xA5) within the specified timeout (0x64: Set to Discharge Mode), until the CU reports that maintenance is finished (bit a = 0).
Retriggering during maintenance is independent on the daycounter value.
The pump then sends one additional discharge request - with byte 3 = 0x00 - to reset the day-counter threshold to the default value (Day Counter).
The pump shall only trigger maintenance when the CU is supplied via mains.
During maintenance, no battery test shall be triggered by the pump.
FCC correction is done independently and with respect to the cut-off voltage aswell during maintenance (Cut-Off Voltage).

Upstream Customer IDs:
41

1.3.2.3.9 [BB000-SysReq-62] Maintenance after Startup

The CU shall evaluate the measured battery voltage after a MCU start up.
If the voltage indicates deep discharged cells (measured voltage <= 3.9 V), the CU shall set the FCC to 2/3 of the last FCC that was valid before the shutdown.
Otherwise the last FCC shall be kept.
The CU shall then charge the battery (refer to Start of Charge Process).
The CU shall set the daycounter to the maximum value, which tells the pump that maintenance is needed (refer to Battery Maintenance).

The CU shall detect, if the start up has been performed after a shutdown.
In this case, the daycounter shall be set to the maximum value, independent on the measured battery voltage.

Remarks:
The battery needs to be charged until End of Charge is detected (End-of-Charge Detection).
The maintenance eventually needs to be triggered by the user.
If the user does not trigger maintenance despite the pump prompting him, the remaining runtime indication should yet be close to the real remaining runtime.
This is why the FCC shall be set to 2/3 of the last FCC for deep discharged batteries.

Upstream Customer IDs:
42

1.3.3 Charge Control and Monitoring

1.3.3.1 Charge Control

1.3.3.1.1 [BB000-SysReq-69] Start of Charge Hysteresis

The CU shall implement the following hysteresis.
The CU shall set an internal flag that inhibits fast charging, when an end of charge is detected.
The CU shall reset this flag, and thus allow fast charging, when the battery has been discharged down to the following value.

Condition
Capacity counter < last FCC - 95 mAh

Remarks:
The value of 95 mAh corresponds to 5% of the nominal FCC.
Also refer to Full Charge Capacity and Capacity Counter.

Upstream Customer IDs:
64

1.3.3.1.2 [BB000-SysReq-46] End-of-Charge Detection

The CU shall detect the battery as fully charged, if one of the following critieria is fulfilled

Effective

V_{max}	If the CU detects the characteristic voltage maximum of NiMH batteries, the CU shall detect the battery as being fully charged.
t_{max}	If the battery has been charged for >= t_{max}, the CU shall detect the battery as being fully charged. $t_{max} = t_{offset} + t_{basis} * (1 - c_{counter} / fcc_{nom})$ t_{offset} = 35 min t_{basis} = 355 min $c_{counter}$ = capacity counter (specified in Capacity Counter) fcc_{nom} = nominal FCC (specified in Full Charge Capacity) The t_{max}-counter shall be stopped, when the charging process is interrupted. The t_{max}-counter shall be reset, when the charging process is started after standby.

After the battery was detected as fully charged, the CU shall switch to Trickle Charge Mode.

Remarks:
For setting of a flag, that indicates a fully charged battery, refer to Start of Charge Hysteresis

Upstream Customer IDs:
20

1.3.3.1.3 [BB000-SysReq-55] Temperature Conditions

The CU shall select the charge mode depending on the following temperature conditions.

Measured battery temperature		Charge Mode	Reference
Min.	Max.		
-	0 °C	Warm-up charging	Warm-Up Charge Mode
0 °C	55 °C	Start fast or trickle charging	Fast Charge Mode, Trickle Charge Mode
65 °C	-	- (Interrupt charging)	-

If no charge mode is applicable, the CU shall wait for the battery to heat up or cool down due to the ambient temperature.

Remarks:
Minimum and maximum ambient operation temperatures are specified in Temperature Ratings
From 55 °C to 65 °C a charging process is not started. However, if this temperature intervall is reached during charging, the charging process is continued until 65 °C, where it is interrupted.
To continue again with the charging process, the CU waits until the measured battery temperature decreases again below 55°C.
For temperature information provided by the communication interface refer to Battery Temperature Monitoring.
For possible ambient temperatures refer to Temperature Ratings.

Effective

Upstream Customer IDs:
12

1.3.3.2 Charge Modes

1.3.3.2.1 [BB000-SysReq-26] Warm-Up Charge Mode

In warm-up charge mode, the CU shall charge the battery with pulses of constant current as specified in Charge Current.

Charge pulse width	1 sec
Charge pause width	20 sec

Remarks:
For temperature conditions for this charge mode refer to Temperature Conditions.

Upstream Customer IDs:

1.3.3.2.2 [BB000-SysReq-25] Fast Charge Mode

In fast charge mode, the CU shall charge the battery with a constant charge current as specified in Charge Current

Upstream Customer IDs:

1.3.3.2.3 [BB000-SysReq-27] Trickle Charge Mode

In trickle charge mode, the CU shall charge the battery with pulses of constant charge current as specified in Charge Current

Charge pulse width	8 sec
Charge pause width	299 sec

Upstream Customer IDs:
21

Effective

1.3.3.3 Charge Monitoring

1.3.3.3.1 [BB000-SysReq-40] Charge Over-Current

The CU shall detect and communicate (in reply to 0x73: Read Status), if the current that is flowing into the battery is to high.

Condition	Bit in 0x73: Read Status	Comment
Measured current into battery > 450 mA for > 300 ms	g = 1	Charge over-current
Else	g = 0	No charge over-current

If the CU detects a charge over-current, charging shall be interrupted for 20 sec.
Then, the CU shall restart the charging process (retry).
In case of a Vin reconnect, the CU shall restart the charging process directly.

Remarks:
Also refer to Charge Current.

Upstream Customer IDs:
12

1.3.3.3.2 [BB000-SysReq-41] Unallowed Discharging

The CU shall detect the following conditions.

Condition	Comment
Measured discharging current >= 30 mA in charging mode	Discharge current detected in charging mode

If the condition is true, the CU shall detect that the battery is being discharged and stop the charging process (switch to discharging mode).

Remarks:
This incident is not communicated via the communication interface.
The charging mode is described in requirement Active Mode; a charging current is not neccessarily flowing in this mode.
Note, that a discharge current might not be detected/measurable, when a charge current is flowing.
Measured current refers to the current measured by the MCU corrected by the average self consumption of the electronic.

Upstream Customer IDs:
12
38

Effective

1.3.3.3.3 [BB000-SysReq-89] Charge Under-Current

The CU shall detect, if the current that is flowing into the battery is to low.

Condition	Comment
Measured current into battery < 350 mA for > 300 ms	Charge under-current
Else	No charge under-current

If the CU detects a charge under-current, charging shall be interrupted for 20 sec.
Then, the CU shall restart the charging process (retry).
In case of a Vin reconnect, the CU shall restart the charging process directly.

Remarks:
This incident is not communicated via the communication interface.
Also refer to Charge Current.

Upstream Customer IDs:

1.3.4 System Integrity Checks

1.3.4.1 [BB000-SysReq-28] NTC Open / Short

The CU shall detect and communicate (in reply to 0x73: Read Status) the following condition.

Condition	Bit in 0x73: Read Status	Comment
NTC Open / Short	n = 1	Defect
Else	n = 0	OK

The CU shall set the charging current to zero until the NTC is detected as OK again.

Remarks:
For the NTC refer to Temperature Measurement

Upstream Customer IDs:

1.3.5 Miscellaneous

1.3.5.1 [BB000-SysReq-30] Power Supply Detection

The CU shall detect the following conditions with a hysteresis.

Condition	Comment
Measured input voltage > 9.8 V	Power supply connected, initial threshold
Measured input voltage < 9.3 V	Power supply disconnected

Upstream Customer IDs:
15

1.3.5.2 [BB000-SysReq-33] Magnetic Field Sensor State

The CU shall detect and communicate (in reply to 0x73: Read Status) the following conditions.

Condition	Bit in 0x73: Read Status
Hall-effect sensor detected the presence of a magnetic field	H = 1
Else	H = 0

Upstream Customer IDs:
11

1.3.5.3 [BB000-SysReq-34] External LED Control

The CU shall enter the LED indication mode, when it does not receive a valid byte from the UART for 1 sec.
At start-up, before communication is detected, the CU shall directly enter the LED indication mode.
The CU shall indicate the following conditions.

Condition	Indication
Warm-up charging is active	Pull TxD to ground permanently
Fast charging is active	
Trickle charging is active	Pull TxD to ground / TxD high impedance (alternate with 1 Hz and 50 % dutycycle)
No charge mode active	TxD high impedance

Remarks:
TxD refers to the pin at the external connector (External Connector).

Upstream Customer IDs:
22

1.3.5.4 [BB000-SysReq-86] Communication Detection

The CU shall detect the following conditions.

Condition	Comment
No valid byte at UART for 2 sec	Communication disconnected
Positive edge at RxD after communication disconnected	Communication reconnected

Remarks:
RxD refers to the pin at the external connector (External Connector).
Refer to Long-Term Storage and Standby for consequences of communication detection.

Upstream Customer IDs:

1.4 Non-Functional Requirements

1.4.1 Hardware and Mechanical Requirements

1.4.1.1 Electrical Ratings

1.4.1.1.1 [BB000-SysReq-16] Input Voltage

The CU shall work as specified inside this specification for the following input voltages.

Parameter	Min.	Typ	Max.	Unit
Input voltage	10	12	16	V

A tolerance of -0.7 V is allowed for the lower limit of the input voltage (Power Supply Detection).

Upstream Customer IDs:
15

1.4.1.1.2 [BB000-SysReq-83] Inrush Current

The inrush current of the CU (pin +Vin) shall be limited to the following value.

Parameter	Max.	Unit
Maximum inrush current	2.25	A

Upstream Customer IDs:
15

1.4.1.1.3 [BB000-SysReq-84] Input Power

The average input power of the CU shall be limited to the following value.

Parameter	Max.	Unit
Average input power (pin +Vin)	3.5	W

Upstream Customer IDs:
15

1.4.1.1.4 [BB000-SysReq-20] Self-Consumption Limits

The average self-consumption of the CU shall, in accordance with the defined system states, be not greater than specified below.

Parameter		Typ.	Max.	Unit
Self-consumption	Standby (no supply voltage and no communication detected)	85	100	µA
	Shutdown	-	10	µA
	Active mode without supply voltage (communication detected)	< 5	7.5	mA

Remarks:
For valid supply voltage range refer to Input Voltage and Power Supply Detection
For the detection of active communication from the pump refer to Long-Term Storage, and Standby
If, in active mode, despite supply voltage and communication are detected as disconnected, a discharge current is detected (Long-Term Storage), the CU remains in active mode.
If, during wake phase in standby, a discharge current is detected (Standby), the CU enters active mode, despite supply voltage and communication are detected as disconnected.

Upstream Customer IDs:
69
26
16

1.4.1.1.5 [BB000-SysReq-21] Charge Current

The CU shall charge the battery with a constant current (continuously or pulsed).
The CU shall control the charge current with the following steady-state accuracy.

Description	Min	Typ	Max	Unit
Charge current (current into battery, DC)	350	400	450	mA

Peak currents shall be < 750 mA and may have a duration of 100 us.
There is no hard requirement on the current ripple.

Remarks:
For over-current error refer to Charge Over-Current.
For under-current error refer to Charge Under-Current.

Upstream Customer IDs:
18

1.4.1.2 Environmental Conditions

1.4.1.2.1 [BB000-SysReq-17] Temperature Ratings

The CU shall work as specified inside this specification inside the following ambient temperature ranges.

Parameter	Min.	Max.	Unit
Ambient temperature for operation	5	65	°C
Ambient temperatre for storage	-20	55	°C

Upstream Customer IDs:
49
50

1.4.1.2.2 [BB000-SysReq-18] Relative Humidity Ratings

The CU shall work as specified inside this specification inside the following ranges for relative humidity.

Parameter	Min.	Max.	Unit
Operation relative humidity (non condensing)	30	90	%
Storage relative humidity (non condensing)	10	90	%

Effective

Upstream Customer IDs:
51
52

1.4.1.2.3 [BB000-SysReq-19] Air pressure - Clearances

The CU shall comply with clearances according to EN 60601-1 (version specified in Documentation).

Applicable clearances shall be determined based on the following air pressure limits.

Parameter	Min.	Max.	Unit
Operation atmospheric air pressure	500	1060	mbar
Storage atmospheric air pressure	500	1060	mbar

Upstream Customer IDs:
53

1.4.1.2.4 [BB000-SysReq-88] Air pressure - Electrolyte Capacitors

Electrolyte capacitors, that are incorporated in the CU, shall be approved by their manufacturers for the following air pressure limits.

Parameter	Min.	Max.	Unit
Operation atmospheric air pressure	500	1060	mbar
Storage atmospheric air pressure	500	1060	mbar

Upstream Customer IDs:
53

1.4.1.3 Mechanical

1.4.1.3.1 [BB000-SysReq-77] Mechanical Dimensions

The mechanical dimensions of the CU (assembled PCB), shall be defined by the following 3D model.

I0688030000 Akkumodul NiMH 2021-07-05.zip
SHA1: 2FA8B2659AA8080BB4EC9E9B71D313F10C1A4619

Remarks:

Effective

In particular, the following features are included in the 3D model.

- *Size of PCB (unassembled)*
- *Height profile (assembled PCB)*
- *Position of the Hall Sensor*
- *Position of the external connector*
- *Position of the battery connector*

Also refer to Pad Size of Connectors.

Upstream Customer IDs:
1 6 5 4 3 2

1.4.1.3.2 [BB000-SysReq-78] Pad Size of Connectors

The pad size for the external and battery connectors of the CU shall be as follows.

Parameter	Value
Pad size on PCB for the external connector	4 mm x 4 mm
Pad size on PCB for the battery connector	4.5 mm x 4.5 mm

Upstream Customer IDs:

1.4.1.3.3 [BB000-SysReq-74] Vibration and Shock

The CU shall be fully functional after the following tests.

Test	Standard	Title
Test Fh	EN 60068-2-64:2008 + A1:2019	Environmental testing - Part 2-64: Tests - Test Fh: Vibration, broadband random and guidance
Test Eb	EN 60068-2-29:1993	Basic environmental testing procedures - Part 2: Tests; test Eb and guidance: Bump

Test	Relevant Specification
Test Fh	- ASD of 10 Hz to 20 Hz; 0,05 g²/Hz, -3 dB/octave and - ASD of 20 Hz to 150 Hz; 0,05 g²/Hz, -3 dB/octave and - an effective total acceleration of 1.6 grms and a test time of 30 minutes
Test Eb	- 1000 shocks (half sine) with a peak acceleration of 15 g and an impulse duration of 6 ms in the vertical direction.

Remarks:

Effective

The requirement comes from EN 1789:2007, 6.4.1 and EN 13718-1:2002, A.1
ASD = acceleration spectral density

Upstream Customer IDs:
54

1.4.1.4 EMC

1.4.1.4.1 [BB000-SysReq-73] EMC standards

The CU shall fulfill the following EMC standards in standalone mode and in application in the pump (Perfusor Space, Type 8713030).

Standard	Title
IEC 60601-1-2:2007	Medical electrical equipment - Part 1-2: General requirements for basic safety and essential performance - Collateral standard: Electromagnetic disturbances - Requirements and tests
IEC 60601-2-24:2012	Medical electrical equipment - Part 2-24: Particular requirements for basic safety and essential performance of infusion pumps and controllers
CISPR 11:2015, modified + A1:2017	Industrial, scientific and medical equipment - Radio-frequency disturbance characteristics - Limits and methods of measurement (*)

(*) The CU shall comply with Class B.

Remarks:
IEC 60601-2-24 requires compliance with IEC 60601-1-2.

Upstream Customer IDs:
55
57
59

1.4.1.4.2 [BB000-SysReq-75] EMC Condition 1

For EMC standards in particular the following condition shall be tested.

- 10 V/m at 80 MHz - 2.5 GHz

Upstream Customer IDs:
56

Effective

1.4.1.4.3 [BB000-SysReq-76] EMC Condition 2

For EMC standards in particular the following condition shall be tested.

- +/- 6 kV external contact-discharge (External Connector)
- +/- 8 kV air-discharge

Upstream Customer IDs:

1.4.1.5 Measurements

1.4.1.5.1 [BB000-SysReq-45] Shunt Resistance Constraint

The shunt resistors used for measuring charge / discharge currents shall not be greater than the specified limit.

Parameter	Max.	Unit
Nominal shunt resistance	50	mΩ

Upstream Customer IDs:
19

1.4.1.5.2 [BB000-SysReq-71] Current Measurement Quantization

The battery current shall be measured with the following quantization.

Parameter	Max.	Unit
Current measurement quantization step width	0.1	mA

Upstream Customer IDs:

1.4.1.5.3 [BB000-SysReq-72] Temperature Measurement

The CU shall measure a temperature by means of an NTC.
The NTC shall be placed directly on the bottom side of the PCB.

The NTC tolerance shall be as follows.

Parameter	Max.	Unit
NTC component tolerance	± 1.78	K

Remarks:
The given tolerance is the NTC component tolerance of the predecessor system.
By means of the temperature measurement, the battery temperature is estimated.

Upstream Customer IDs:
67
68

1.4.2 Software Requirements

1.4.2.1 [BB000-SysReq-66] Software Version Number

The CU shall report the software version number in reply to 0x76: Read Version
The decimal value of byte 3 in the reply shall have to following format.

value = version number * 100

The version number shall start at 1.60 for mass production.
The version number shall not exceed 1.79.

Remarks:
Version numbers starting from 1.80 are reserved for another system (LiIon).
Latest version number of the old NiMH system is 1.52.

Upstream Customer IDs:

1.4.2.2 In-System Programming

1.4.2.2.1 [BB000-SysReq-22] Bootloader and Service Mode

The CU software shall provide a bootloader to provide the ability to load a new SW image.
The bootloader shall be accessible via the communication interface (Communication Interface).

The CU shall support the following activation sequence:

- “Extended Mode” command
- “Service Mode” command
- “Update” command

In the service mode the bootloader shall be able to load the new SW image.

Remarks:
The update will only be started by a service employee and not by the pump.

The commands for the activation sequence are defined in the appendices (communication protocol). Behavior within the bootloader and relevant commands are specified in the document "Bootloader Specification (SpaceB BMS)" [BSP].

Upstream Customer IDs:
14

1.4.2.2.2 [BB000-SysReq-57] Error Detection Code

The bootloader shall verify the SW image(s) integrity with an error detection code (e.g. CRC).

The system shall not execute SW images who's integrity wasn't verified successfully by this method.

Upstream Customer IDs:

1.4.2.3 Communication Interface

1.4.2.3.1 [BB000-SysReq-38] Communication Interface Configuration

The UART for the communication interface shall be configured as follows.

Baud rate:	19200 Bd
Data bits:	8
Parity bits:	no
Stop bits:	1

Upstream Customer IDs:
46

1.4.2.3.2 [BB000-SysReq-64] Physical Communication Interface

The CU shall provide an open drain/collector output at the TxD pin.
The idle bus-voltage is "high" (non-inverted communication over TTL levels).

During external LED control, the CU shall provide a current sink through the TxD pin for currents that are < 100 mA.

Remarks:
For LED indication refer to External LED Control and Communication Interface
The idle bus-voltage is high, but note that the TxD pin can only be switched to high-impedance or low by the CU.

Upstream Customer IDs:

1.4.2.3.3 [BB000-SysReq-54] Battery Data

The CU shall provide basic system information in reply to 0x76: Read Version.

Parameter	Value
Status bit **T**	**0** (NiMH)
Nominal capacity	**19** [100mAh]

Remarks:
Nominal capacity defined in Full Charge Capacity

Upstream Customer IDs:
44
45

1.4.2.3.4 [BB000-SysReq-39] Response Timing Constraint

After the receipt of a valid byte at the RxD MCU pin, the CU shall send the first byte of the answer within 50 ms.

Remarks:
This also holds for the answer to invalid commands.

Upstream Customer IDs:
46

1.4.3 Safety

1.4.3.1 [BB000-SysReq-79] UL Listed Materials

The following components for the CU shall be UL listed.

- Base material for the PCB
- Fuses

Remarks:
The base material for the PCB will be FR4.

Upstream Customer IDs:
62
7

Effective

1.4.3.2 [BB000-SysReq-87] Documentation

The supplier shall provide the information, that is demanded by the customer in order to declare the CU compliant with the following standards / regulations, according to the Development Interface Agreement [DIA].
The listed standards do not incorporate additional requirements, unless explicitly stated inside this specification, and may solely serve an informational purpose.

Standard / Regulation	Title
EN ISO 14971:2019	Medical devices – Application of risk management to medical devices
EN ISO 13485:2016	Medical devices – Quality management systems – Requirements for regulatory purposes
EN 62304:2006+Cor.:2008+A1:2015	Medical device software - Software life-cycle processes
EN 60601-1:2006+A1:2013	Medical electrical equipment – Part 1: General requirements for basic safety and essential performance
EN 60601-2-24:2015	Medical electrical equipment - Part 2-24: Particular requirements for basic safety and essential performance of infusion pumps and controllers
21 CFR Part 820	Medical Devices - Quality System Regulation

Remarks:
The customer has classified the CU as class B (EN 62304).

Upstream Customer IDs:
62
61

2 [BB000-CMP-1] System Requirements - Appendices

Upstream Customer IDs:

2.1 [BB000-SET-10] Communication Protocol

Upstream Customer IDs:

Effective

2.1.1 Introduction

2.1.1.1 Overview

2.1.1.2 Acronyms and Abbreviations

chs	Checksum
CU	Charging Unit

2.1.2 Communication Protocol

2.1.2.1 [BB000-Comm-9] Invalid Command Behavior

The CU shall detect a received command as invalid, in any of the following conditions.

- Byte 0 not equal to 0x3F ('?') or 0x21 ('!')
- Unsupported command ID (e.g. operation command in service mode)
- Non-valid message data (invalid request)
- Corrupted message (checksum error)

Testercommands shall not be supported outside of tester mode

After receiving the expected number of bytes, the CU shall switch off RxD.
Thus, further bytes shall be ignored, until the CU has sent the answer to the received command.
The CU shall determine the expected number of bytes from byte 1 of the received command.

The CU shall answer to an invalid command with a defined error message as follows (independent on the number of received bytes).

Direction	CU → Comm host	
Byte 0	0x3F	'?'
Byte 1	Return byte 1 received by host	
Byte 2	0	
Byte 3	0	
Byte 4	*chs* (Checksum)	

Remarks:
A too short command by the communication host may result in a random answer (byte 1) by the CU.

Upstream Customer IDs:
46

2.1.2.2 [BB000-Comm-10] Checksum

The checksum shall be calculated in the following manner:

chs = 0 - (byte_1 + byte_2 + byte_3 + byte_4)

Upstream Customer IDs:
46

2.1.3 Operation Commands

2.1.3.1 [BB000-Comm-1] Command Structure

Each command consists of two messages, one message sent from the communication host to the CU and an answer send from the CU back to the communication host.

Each message (communication host to CU and vice versa) shall be five bytes long and structured as follows.

Direction	Comm host → CU / CU → comm host
Byte 0	Write (*0x21* / '!') or read (*0x3F* / '?') command
Byte 1	Command ID
Byte 2	Message data
Byte 3	
Byte 4	Checksum (*chs*)

Remark:
Bytes are ordered from MSB (bit 7) to LSB (bit 0).

Upstream Customer IDs:
46

2.1.3.2 [BB000-Comm-2] 0x73: Read Status

The *Read Status* command shall provide statuses, alarms and warnings from the Charging Unit to the communication host.

Direction	Comm host → CU	
Byte 0	0x3F	'?'
Byte 1	0x73	's'
Byte 2	0	-
Byte 3	0	-
Byte 4	*chs*	

Direction	CU → Comm host		Reference
Byte 0	0x21	'!'	-
Byte 1	0x73	's'	-
Byte 2	Bit 7: a	reserved (0)	-
	Bit 6: w	reserved (0)	-
	Bit 5: s	Charging enabled via serial command (1 = enabled, 0 = disabled)	0x64: Set to Discharge Mode
	Bit 4: u	Vmax criterion	End-of-Charge Detection
	Bit 3: T	reserved (0)	
	Bit 2: t	Time criterion	End-of-Charge Detection
	Bit 1: H	Hall sensor status (1 = active, 0 = inactive)	Magnetic Field Sensor State
	Bit 0: c	Charging status (1 = not trickle charging, 0 = trickle charging)	Battery Function Test - Trickle Charge
Byte 3	Bit 7: n	Temperature sensor error	NTC Open / Short
	Bit 6: h	Temperature not in valid region (hot, cold)	Temperature Conditions
	Bit 5: r	Battery short	Battery Function Test - Charging
	Bit 4: d	Battery volt. deep / under-voltage	Battery Deep Discharged
	Bit 3: m	Battery volt. at minimum / fully discharged	Battery Maintenance
	Bit 2: v	Battery volt. over limit / over-voltage	Battery Function Test - Charging
	Bit 1: i	Discharge over-current	Discharge Over-Current
	Bit 0: g	Charge over-current	Charge Over-Current
Byte 4	*chs*		-

Upstream Customer IDs:
46

2.1.3.3 [BB000-Comm-3] 0x63: Read Capacity

The *Read Capacity* command shall provide the by the Charging Unit estimated current capacity or the last full charged capacity.

Direction	Comm host → CU	
Byte 0	0x3F	'?'
Byte 1	0x63	'c'
Byte 2	0 / 0xA5	0x00 = current capacity, 0xA5 = last full charged capacity
Byte 3	0	-
Byte 4	*chs*	

Direction	CU → Comm host		Reference
Byte 0	0x21	'!'	
Byte 1	0x63	'c'	
Byte 2	Low-byte	Current capacity or last full charged capacity in [mAh], unsigned integer	Remaining Capacity Estimation,Full Charge Capacity Estimation
Byte 3	High-byte		
Byte 4	*chs*		

Upstream Customer IDs:
46

2.1.3.4 [BB000-Comm-4] 0x76: Read Version

The *Read Version* command shall provide the software version and further battery info.

Direction	Comm host → CU	
Byte 0	0x3F	'?'
Byte 1	0x76	'v'
Byte 2	0	-
Byte 3	0	-
Byte 4	*chs*	

Direction	CU → Comm host		Reference
Byte 0	0x21	'!'	
Byte 1	0x76	'v'	
Byte 2	SW version	Value / 100 = version number, e.g. 160 → version number 1.60	Software Version Number
Byte 3	Bit 7: T	Battery type (0 = NiMH, 1 = LiIon)	Battery Data
	Bit 6: b	Beta version (0 = no, 1 = yes)	-
	Bit 5 … Bit 0	Nominal capacity [100mAh], unsigned integer	Battery Data
Byte 4	*chs*		

Upstream Customer IDs:
46

2.1.3.5 [BB000-Comm-5] 0x64: Set to Discharge Mode

This command shall trigger the CU to stop charging.
The CU shall automatically leave this discharge mode after 15 sec.
A newly received command shall restart the timer again, i.e. extending the pause.

Direction	Comm host → CU		Reference
Byte 0	0x21	'!'	-
Byte 1	0x64	'd'	-
Byte 2	0 / 0xA5	0x00 = daycounter reset at 30 min rem. time, 0xA5 = daycounter reset at cut-off voltage, This setting is kept even after leaving the discharge mode after 15 sec!	Battery Maintenance, Cut-Off Voltage
Byte 3	0	-	-
Byte 4	*chs*		-

Direction	CU → Comm host		Reference
Byte 0	0x21	'!'	-
Byte 1	0x64	'd'	-
Byte 2	Bit 7: s	Confirm that charging is disabled (0 = disabled)	Battery Maintenance
	Bit 6: a	Battery maintenance status (0 = finished, 1 = running)	Battery Maintenance
	Bit 5 … Bit 0 = 0x00	Reserved	-
Byte 3	0	N/A	-
Byte 4	*chs*		-

Remarks:
With this command the CU is triggered to stop charging. Thus, bit s is always set to zero to confirm, that charging is disabled (compatibility with old system).
Bit s can also be found in 0x73: Read Status.
This command is used by the pump in Battery Function Test - Trickle Charge and Battery Maintenance.

Upstream Customer IDs:
46
32

2.1.3.6 [BB000-Comm-6] 0x74: Read Day Counter

The *Read Day Counter* command shall provide the value of the day counter.

Direction	Comm host → CU	
Byte 0	0x3F	'?'
Byte 1	0x74	't'
Byte 2	0	-
Byte 3	0	-
Byte 4	*chs*	

Direction	CU → Comm host		Reference
Byte 0	0x21	'!'	-
Byte 1	0x74	't'	-
Byte 2	Low-byte	Day counter value	Day Counter
Byte 3	0	-	-
Byte 4	*chs*		-

Upstream Customer IDs:
46

2.1.3.7 [BB000-Comm-7] 0x72: Read Remaining Time

The *Read Remaining Time* command shall provide the remaining time in minutes until fully discharged.

Direction	Comm host → CU	
Byte 0	0x3F	'?'
Byte 1	0x72	'r'
Byte 2	0	-
Byte 3	0	-
Byte 4	*chs*	

Direction	CU → Comm host		Reference
Byte 0	0x21	'!'	-
Byte 1	0x72	'r'	-
Byte 2	Low-byte	Remaining runtime [min], unsigned integer	Remaining Runtime Estimation
Byte 3	High-byte		
Byte 4	*chs*		-

Upstream Customer IDs:
46

Effective

2.1.3.8 [BB000-Comm-30] 0x72: Set Remaining Time

The *Set Remaining Time* command shall set the estimate of the remaining runtime for debugging purposes.
This command shall not be used in normal pump operation, where it leads to undefined system behavior.

The CU shall also recalculate the capacity counter (and remaining capacity) according to the requested set value of the remaining runtime and following the caclulation specification from Remaining Runtime Estimation and Remaining Capacity Estimation.

If the resulting value for the capacity counter is greater than the nominal full charge capacity (Full Charge Capacity), the capacity counter shall be set to the nominal full charge capacity instead.
Else, if the resulting value for the capacity counter is greater than the last full charge capacity (Full Charge Capacity Estimation), the capacity counter shall be set to the last full charge capacity instead.

The estimation error (described in Full Charge Capacity Estimation) shall be set to zero.

The CU shall evaluate Start of Charge Hysteresis for the new capacity counter value.
The CU shall consider the battery to be empty if the RR <= zero, otherwise not empty.

Direction	Comm host → CU	
Byte 0	0x21	'!'
Byte 1	0x72	'r'
Byte 2	Low-Byte	Remaining runtime [min], unsigned integer
Byte 3	High-Byte	
Byte 4	*chs*	

The CU shall return the commanded value, if the remaining runtime has been set.
The CU shall return the actual remaining runtime, if the remaining runtime has not been set.

Direction	CU → Comm host	
Byte 0	0x21	'!'
Byte 1	0x72	'r'
Byte 2	Low-byte	Remaining runtime [min], unsigned integer
Byte 3	High-byte	
Byte 4	*chs*	

Remarks:
This command is used in analysis tools and supported for compatibility to the old system.

Upstream Customer IDs:

2.1.3.9 [BB000-Comm-8] 0x61: Get Accutest Result

The *Get Accutest Result* command shall activate the battery test (Battery Function Test - Trickle Charge) and provide the battery test result.

Direction	Comm host → CU	
Byte 0	0x21	'!'
Byte 1	0x61	'a'
Byte 2	0	-
Byte 3	0	-
Byte 4	*chs*	

Direction	CU → Comm host	
Byte 0	0x21	'!'
Byte 1	0x61	'a'
Byte 2	Bit 7: i	Current (1 = too low, 0 = ok)
	Bit 6: u	Voltage (1 = too low, 0 = ok)
	Bit 5 ⋯ Bit 2 = 0x0	Reserved
	Bit 1: e	Battery test enabled (1 = enabled)
	Bit 0: p	Battery test (1 = passed / 0 = not passed)
Byte 3	Bit 7 ⋯ Bit 0 = 0x00	Reserved
Byte 4	*chs*	

Upstream Customer IDs:
46

2.1.3.10 [BB000-Comm-20] 0x31: Get Serial Number - Low Word

The *Get Serial Number - Low Word* command shall provide the lower two bytes of the serial number.

Direction	Comm host → CU	
Byte 0	0x3F	'?'
Byte 1	0x31	'1'
Byte 2	0	-
Byte 3	0	-
Byte 4	*chs*	

Direction	CU → Comm host	
Byte 0	0x21	'!'
Byte 1	0x31	'1'
Byte 2	Low-byte	Serial number - Byte 0
Byte 3	High-byte	Serial number - Byte 1
Byte 4	*chs*	

Upstream Customer IDs:
66

2.1.3.11 [BB000-Comm-19] 0x32: Get Serial Number - High Word

The *Get Serial Number - High Word* command shall provide the upper two bytes of the serial number.

Direction	Comm host → CU	
Byte 0	0x3F	'?'
Byte 1	0x32	'2'
Byte 2	0	-
Byte 3	0	-
Byte 4	*chs*	

Direction	CU → Comm host	
Byte 0	0x21	'!'
Byte 1	0x32	'2'
Byte 2	Low-byte	Serial number - Byte 2
Byte 3	High-byte	Serial number - Byte 3
Byte 4	*chs*	

Upstream Customer IDs:
66

2.1.4 Tester Commands

2.1.4.1 [BB000-Comm-23] 0xA1: Enable Tester Mode

This command shall enable the tester mode.
If a tester command is sent without enabled tester mode, the command shall be detected as invalid.

Direction	Comm host → CU	
Byte 0	0x21	'!'
Byte 1	0xA1	Enable tester mode
Byte 2	0x5A	Password low byte
Byte 3	0x45	Password high byte
Byte 4	*chs*	

The CU shall return the following answer.

Direction	CU → Comm host	
Byte 0	0x21	'!'
Byte 1	0xA1	Confirmation - tester mode enabled
Byte 2	0x00	-
Byte 3	0x00	-
Byte 4	*chs*	

Upstream Customer IDs:

2.1.4.2 [BB000-Comm-24] 0xA2: Disable Tester Mode

This command shall disable the tester mode.

Direction	Comm host → CU	
Byte 0	0x21	'!'
Byte 1	0xA2	Disable tester mode
Byte 2	0x00 - 0xFF	-
Byte 3	0x00 - 0xFF	-
Byte 4	*chs*	

The CU shall return the following answer.

Direction	CU → Comm host	
Byte 0	0x21	'!'
Byte 1	0xA2	Confirmation - tester mode disabled
Byte 2	0x00	-
Byte 3	0x00	-
Byte 4	*chs*	

Upstream Customer IDs:

2.1.4.3 [BB000-Comm-31] 0x41: Get AD Values

If the software receives the following command

Direction	Comm host → CU	
Byte 0	0x3F	'?'
Byte 1	0x49	Fix byte for tester mode commands
Byte 2	0x41	Get AD values
Byte 3	0x00	-
Byte 4	0x00	-
Byte 5	*chs*	

the software shall return the following answer

Direction	CU → Comm host	
Byte 0	0x21	'!'
Byte 1	0x41	Get AD values
Byte 2	0x00 - 0xFF	*Low-byte*: VinAD
Byte 3	0x00 - 0xFF	*High-byte*: VinAD
Byte 4	0x00 - 0xFF	*Low-byte*: VBatAD
Byte 5	0x00 - 0xFF	*High-byte*: VBatAD
Byte 5	0x00 - 0xFF	*Low-byte*: VCurrentAD [1]
Byte 5	0x00 - 0xFF	*High-byte*: VCurrentAD [1]
Byte 5	0x00 - 0xFF	*Low-byte*: VNtcAD
Byte 5	0x00 - 0xFF	*High-byte*: VNtcAD
Byte 5	*chs*	

Remark:
[1] The returned AD value is the the result of MAX($AD_{ChargeCurrent}$, $AD_{DischargeCurrent}$).

Upstream Customer IDs:

2.1.4.4 [BB000-Comm-25] 0x46: Get Flags

In reply to this command, the CU shall provide status bits in the returned answer.

Direction	Comm host → CU	
Byte 0	0x3F ('?')	-
Byte 1	0x49	Fix byte for test commands
Byte 2	0x46	Get Flags CMD
Byte 3	0x00	Reserved

Byte 4	0x00	Reserved
Byte 5	*chs*	

The CU shall return the following answer.

Direction	Comm host → CU	
Byte 0	0x21 ('!')	-
Byte 1	0x46	Get Flags CMD
Byte 2	0x00 - 0xFF	*Low-byte*: Error status
Byte 3	0x00 - 0xFF	*High-byte*: Error status
Byte 4	0x00 - 0xFF	Charge state
Byte 5	0x00 - 0xFF	Reserved
Byte 6	0x00 - 0xFF	Reserved
Byte 7	0x00 - 0xFF	Reserved
Byte 8	0x00 - 0xFF	Reserved
Byte 9	0x00 - 0xFF	Reserved

The answer shall contain the following bits for Byte 2 and 3.

Byte 2: Error status Low Byte		
Bit 0	b0/1	Charger overvoltage
Bit 1	b0/1	Temperature sensor fault
Bit 2	b0/1	Current sensor fault
Bit 3	b0/1	Internal system fault
Bit 4	b0/1	Statemonitor parameter fault
Bit 5	b0/1	Battery overvoltage
Bit 6	b0/1	Battery short circuit
Bit 7	b0/1	Charge overcurrent

Byte 3: Error status High Byte		
Bit 0	b0/1	Charge undercurrent
Bit 1	b0/1	Production data faulty
Bit 2	b0	Reserved
Bit 3	b0	Reserved
Bit 4	b0	Reserved
Bit 5	b0	Reserved
Bit 6	b0	Reserved
Bit 7	b0	Reserved

The answer shall contain the following information in Byte 4.

Byte 4: Charge states		
Byte 4	0x00	Initialization
	0x01	Idle
	0x02	Warm-up charging
	0x03	Fast charging
	0x04	Trickle charging
	0x05	Charging pause

Upstream Customer IDs:

2.1.5 Service Commands

2.1.5.1 [BB000-Comm-27] 0x65: Enable Extended Mode

The communication host may send this command prior to 0x66: Enable Service Mode.

Direction	Comm host → CU	Description
Byte 0	0x21	'!'
Byte 1	0x65	'e' - Enable extended mode
Byte 2	0x00 / 0x80	0x00: Extended mode disable 0x80: Extended mode enable
Byte 3	0x00	-
Byte 4	*chs*	-

The CU shall return the following answer.

Direction	CU → Comm host	Description
Byte 0	0x21	'!'
Byte 1	0x65	'e' - Confirmation - Extended mode enabled
Byte 2	0x00 / 0x80	Return byte 2 of received command
Byte 3	0x00	-
Byte 4	*chs*	-

Remarks:
This command is supported to be compatible to the customer service tool.

Upstream Customer IDs:

Effective

2.1.5.2 [BB000-Comm-26] 0x66: Enable Service Mode

This command shall enable the service mode (Bootloader and Service Mode).

Direction	Comm host → CU	
Byte 0	0x21	'!'
Byte 1	0x66	'f' - Enable service mode
Byte 2	0xBE	Password low byte
Byte 3	0xEF	Password high byte
Byte 4	*chs*	-

The CU shall return the following answer.

Direction	CU → Comm host	
Byte 0	0x21	'!'
Byte 1	0x66	'f' - Confirmation - Service mode enabled
Byte 2	0x00	-
Byte 3	0x00	-
Byte 4	*chs*	-

Upstream Customer IDs:

2.1.5.3 [BB000-Comm-28] 0x66: Update

The CU shall be able to receive the following frame.

Direction	Comm host → CU	
Byte 0	0x21	'!'
Byte 1	0x66	'f' - enable firmware update
Byte 2	0xB0	Password low byte
Byte 3	0x01	Password high byte
Byte 4	*chs*	-

The CU shall return the following answer.

Direction	CU → Comm host	
Byte 0	0x21	'!'
Byte 1	0x66	'f' - Confirmation - Firmware update enabled
Byte 2	0x00	-
Byte 3	0x00	-

Effective

Effective

Byte 4	*chs*	-

Upstream Customer IDs:

Effective

Title: Space Battery NiMH - Charging Unit System Requirement Specification Initiator: Ruediger ? Haupt

This document is signed electronically in compliance with the B. Braun electronic signature policies and procedures by following persons:

UserName: Haupt, Ruediger (hauprude)
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